Toxicity-Screened High-Frequency Market-Making: An Optimal Control Framework with Empirical Validation

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Abstract

We evaluate whether real-time toxicity screening can improve market-making profitability on NYSE equity markets. Extending the Avellaneda-Stoikov optimal control framework, we develop an online SGD logistic model for order flow toxicity estimation using 15 microstructure features (5 order-book ratios, 3 temporal dynamics, 7 structural features) and compare three filter variants–SGD logistic, walk-forward, and EWMA adaptive threshold–backtested on 74 GB of NYSE XDP Integrated Feed data (1.27 billion messages, August 22, 2023) against an unscreened baseline. All three toxicity-aware filters achieve positive P&L: SGD logistic +\$1,266, walk-forward +\$1,353, EWMA +\$277, on approximately 2,223 fills with +\$0.57 per fill realized. The E[PnL] quoting filter must use the base spread rather than the toxicity-adjusted spread, because fills near the NBBO capture approximately base spread regardless of the wider quoted spread. Inventory risk is controlled by a quadratic penalty ($\gamma = 2 \times 10^{-7}$) that suppresses quoting at ∼75 shares inventory, with periodic unwind crosses for excess positions. The toxicity score has Spearman $\rho = 0.124$ with realized adverse selection ($p < 10^{-10}$, $n = 30{,}816$), with cancellation rate alone accounting for $\rho = 0.101$. The dominant mechanism is loss avoidance, not alpha generation. Online SGD learning does not improve over static weight initialization, confirming the initialization is near-optimal. The inventory risk penalty γ is the critical parameter: $5 \times 10^{-8} \to -\$32\text{K}$, $10^{-7} \to -\$4.5\text{K}$, $2 \times 10^{-7} \to +\$1.3\text{K}$. Results are limited to a single trading day.

1 Introduction and Problem Formulation

1.1 Market-Making as Stochastic Optimal Control

We formulate market-making in a limit order book (LOB) as a discrete-time stochastic optimal control problem over $[0, T]$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space with filtration $\{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions. The decision process evolves on a discrete time grid $\mathcal{T} = \{t_0, t_1, \ldots, t_N\}$ with $t_k = k\delta$ for fixed step $\delta > 0$.

1.2 State Variables and Control Space

The system state at time t_k is $(Q_{t_k}, P_{t_k}, S^{bid}_{t_k}, S^{ask}_{t_k}, \mathcal{B}_{t_k}, \mathcal{A}_{t_k}, \Theta_{t_k})$ comprising inventory $Q_{t_k} \in \mathbb{Z}$, mid-price P_{t_k}, best bid/ask, full order book state (price-volume pairs on each side), and a d-dimensional microstructure feature vector Θ_{t_k}.

The controls are: half-spread $\Delta_{t_k} \in [\Delta_{\min}, \Delta_{\max}]$, binary quoting indicators $\xi^{bid}_{t_k}, \xi^{ask}_{t_k} \in \{0, 1\}$, and quote sizes $V^{bid}_{t_k}, V^{ask}_{t_k} \in \mathbb{Z}_+$.

1.3 Related Work

Avellaneda and Stoikov (2008) derive closed-form optimal quotes under exponential utility; Guéant, Lehalle, and Fernandez-Tapia (2012) extend this to market order arrival dynamics. The adverse selection framework originates with Glosten and Milgrom (1985) and Kyle (1985), who formalize informed-trader adverse selection; Easley et al. (1996) introduce the PIN model. Our toxicity model is a real-time, high-frequency analog of PIN using order flow microstructure. On the empirical side, Menkveld (2013) documents HFT market-maker behavior, Brogaard, Hendershott, and Riordan (2014) analyze HFT information content, and Cartea and Jaimungal (2015) incorporate latency and queue priority. Easley, López de Prado, and O'Hara (2012) introduce VPIN from trade volume aggregation; our approach instead computes toxicity from order book dynamics (cancellation rates, imbalance), enabling lower-latency estimation. We extend Avellaneda-Stoikov with real-time toxicity adjustment, provide implementable algorithms, and validate on 74 GB of NYSE XDP data.

2 Data Source

2.1 NYSE XDP Integrated Feed

We use the NYSE Integrated Feed (TAPE A) delivered via the XDP binary protocol (v2.3a). Five message types support order book reconstruction: ADD (100), MODIFY (101), DELETE (102), EXECUTE (103), and REPLACE (104), each containing order ID, symbol index, price, volume, side, and nanosecond timestamp.

2.2 Data Characteristics

The data comprises a complete trading day (August 22, 2023): 6.5 hours (9:30 AM–4:00 PM ET), 74GB across 122 time-sequential PCAP files, 1.27 billion XDP messages (peak >500K msgs/sec), 289 million packets, ∼3,500 NYSE-listed securities. Individual 64-bit order IDs enable precise book reconstruction and queue position estimation.

2.3 Market Environment

August 22, 2023: moderate volatility (VIX 16–17, S&P 500 ∼4,387), fed funds 5.25–5.50%, Jackson Hole later that week. Characteristic intraday patterns present (elevated open volume, quiet midday, MOC-driven close). Single-day test case; multi-day validation remains a priority (Section 12.2).

3 Mathematical Foundations and Assumptions

3.1 Market Microstructure Assumptions

Assumption 3.1 (Limit Order Book Structure)**.** Prices lie on a tick grid $p \in \tau\mathbb{Z}$ ($\tau > 0$). The book has bid and ask sides $\mathcal{B}_t = \{(p_i^{bid}, v_i^{bid})\}_{i=1}^{L_b}$, $\mathcal{A}_t = \{(p_i^{ask}, v_i^{ask})\}_{i=1}^{L_a}$, with $S_t^{bid} = \max\{p : (p, v) \in \mathcal{B}_t\}$, $S_t^{ask} = \min\{p : (p, v) \in \mathcal{A}_t\}$, and $P_t = (S_t^{bid} + S_t^{ask})/2$.

Assumption 3.2 (Price Process)**.** The mid-price follows a jump-diffusion semimartingale:

$$dP_t = \mu_t dt + \sigma_t dW_t + \sum_{i:\tau_i \le t} \Delta P_{\tau_i} \tag{1}$$

Assumption 3.3 (Order Flow Dynamics)**.** Additions $\{N_t^{add}\}$, cancellations $\{N_t^{cancel}\}$, and executions $\{N_t^{exec}\}$ follow point processes with book-state-dependent intensities λ_t^{add}, λ_t^{cancel}, λ_t^{exec}.

Assumption 3.4 (Adverse Selection)**.** A latent toxicity $\theta_t \in [0, 1]$ governs expected adverse price movement conditional on a fill:

$$\mathbb{E}[\Delta P_{t+\delta} \mid \text{fill at } t, \theta_t] = \begin{cases} -\mu_{adv}\theta_t & \text{if we buy (bid fill)} \\ +\mu_{adv}\theta_t & \text{if we sell (ask fill)} \end{cases} \tag{2}$$

where $\mu_{adv} > 0$ is the adverse movement per unit toxicity.

Assumption 3.5 (Fill Probability Model)**.** The fill probability $P_{\text{fill}}(p, v, \Delta_t, \mathcal{B}_t, \mathcal{A}_t)$ depends on queue position, spread, flow intensity, and latency. For tractability: $P_{\text{fill}} \approx \bar{P}_{\text{fill}} \cdot \mathbf{1}\{p \text{ at or better than best bid/ask}\}$ where $\bar{P}_{\text{fill}} \in (0, 1)$ is calibrated.

Assumption 3.6 (Information Structure)**.** The market maker observes $(\mathcal{B}_t, \mathcal{A}_t)$ up to L levels, historical flow $\{N_s^{add}, N_s^{cancel}, N_s^{exec} : s \leq t\}$, own inventory Q_t, and P&L Π_t. Counterparty identity, hidden orders, and future flow are unobserved.

Assumption 3.7 (Execution Model)**.** Fills occur at our quoted price (price-time priority) with size $\min\{V_t, V_{market}\}$, after latency $\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2)$ microseconds, with queue position ahead $Q_{ahead} \sim \mathcal{N}(\mu_q, \sigma_q^2)$ truncated at zero.

4 Toxicity Model: Statistical Foundation

4.1 Definition and Observable Proxies

Definition 4.1 (Toxicity Score)**.** The toxicity score $T_t \in [0, 1]$ is a $\mathcal{F}_t$-measurable random variable representing the conditional probability that the next trade originates from an informed trader, given the current microstructure state:

$$T_t = \mathbb{P}(\text{informed trade} \mid \mathcal{F}_t) \tag{3}$$

Since T_t is latent, we construct a proxy $\hat{T}_t$ from observable microstructure features via online SGD logistic regression.

4.2 Microstructure Feature Vector

The model uses a 15-dimensional feature vector in three groups, summarized in Table 1.

The first group consists of five order book ratios averaged across the top $L = 3$ bid and ask levels, each normalized by total activity (adds + cancels). The cancel ratio C_t, computed over a rolling window $[t - W, t]$, is:

$$C_t = \frac{N_t^{cancel} - N_{t-W}^{cancel}}{(N_t^{add} + N_t^{cancel}) - (N_{t-W}^{add} + N_{t-W}^{cancel})} \tag{4}$$

The remaining four order book features–ping ratio R_t^{ping} (orders < 10 shares), odd lot ratio R_t^{odd} (volume not divisible by 100), precision ratio R_t^{prec} (sub-penny pricing, $|p - \text{round}(100p)/100| > 0.0001$), and resistance ratio R_t^{res} (orders at psychological levels such as \$X.01, \$X.05, \$X.95, \$X.98, \$X.99)–are each computed as the count of qualifying events divided by total activity.

The second group comprises three temporal features from per-symbol circular buffers: trade flow imbalance $I_t^{flow} = (V_{buy} - V_{sell})/(V_{buy} + V_{sell})$ over the last 100 executions, spread change rate $\Delta s_t = (s_t - s_{t-N})/s_{t-N}$ over the last $N = 50$ spread observations, and price momentum $m_t = (P_t - P_{t-N})/P_{t-N}$ over the last $N = 50$ mid-price observations.

The third group consists of seven structural statistics computed across the top $L = 3$ levels: cancel volume intensity CVI_t (volume cancelled / volume added), top-of-book concentration TBC_t (best-level quantity as a fraction of total visible depth, averaged across sides), depth imbalance $\mathrm{DI}_t = (V_t^{bid,total} - V_t^{ask,total})/(V_t^{bid,total} + V_t^{ask,total})$ using full visible depth, level asymmetry $\mathrm{LA}_t = (L_t^{bid} - L_t^{ask})/(L_t^{bid} + L_t^{ask})$, absolute trade imbalance $|\mathrm{TFI}_t|$, large order ratio LOR_t (orders > 200 shares / total events), and normalized spread $\mathrm{NS}_t = s_t/P_t$.

Group	**Feature**	**Symbol**
Order book (5)	Cancel ratio	C_t
	Ping ratio	R_t^{ping}
	Odd lot ratio	R_t^{odd}
	Precision ratio	R_t^{prec}
	Resistance ratio	R_t^{res}
Temporal (3)	Trade flow imbalance	I_t^{flow}
	Spread change rate	Δs_t
	Price momentum	m_t
Structural (7)	Cancel volume intensity	CVI_t
	Top-of-book concentration	TBC_t
	Depth imbalance	DI_t
	Level asymmetry	LA_t
	Absolute trade imbalance	$\lvert\mathrm{TFI}_t\rvert$
	Large order ratio	LOR_t
	Normalized spread	NS_t

Table 1: The 15 microstructure features grouped by category. All order book features are normalized by total activity (adds + cancels) across the top $L = 3$ levels.

The complete feature vector is:

$$\mathbf{x}_t = (C_t, R_t^{ping}, R_t^{odd}, R_t^{prec}, R_t^{res}, I_t^{flow}, \Delta s_t, m_t, \mathrm{CVI}_t, \mathrm{TBC}_t, \mathrm{DI}_t, \mathrm{LA}_t, |\mathrm{TFI}_t|, \mathrm{LOR}_t, \mathrm{NS}_t) \in \mathbb{R}^{15} \tag{5}$$

Order book imbalance (OBI), used separately in the quoting policy (Section 5) but not input to the toxicity score:

$$\mathrm{OBI}_t = \frac{V_t^{bid} - V_t^{ask}}{V_t^{bid} + V_t^{ask}} \in [-1, 1] \tag{6}$$

where $V_t^{bid} = \sum_{i=1}^{L} v_i^{bid}$ and $V_t^{ask} = \sum_{i=1}^{L} v_i^{ask}$ aggregate visible volumes across L book levels.

4.3 SGD Toxicity Model

The predicted toxicity is:

$$\hat{T}_t = \sigma(\mathbf{w}^\top \tilde{\mathbf{x}}_t + b) \tag{7}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the sigmoid function, $\tilde{\mathbf{x}}_t$ is the z-score normalized feature vector, and $(\mathbf{w}, b)$ are the model parameters. Normalization uses Welford's online algorithm: each feature

is standardized as $\tilde{x}_i = (x_i - \hat{\mu}_i)/\hat{\sigma}_i$ where $\hat{\mu}_i$ and $\hat{\sigma}_i$ are running mean and standard deviation estimates updated on each fill observation.

The initial weights $\mathbf{w}_0 = (0.4, 0.2, 0.15, 0.15, 0.1, 0, \ldots, 0)$ and $b_0 = 0$ assign prior mass to the five order book features proportional to their expected informativeness; temporal and structural features start at zero (learned from data). During warm-up ($N_w = 2$ fills), raw features are used before normalization statistics stabilize.

After each fill at time t, we observe the binary adverse selection outcome $y_t = \mathbf{1}\{\text{adverse move} > \epsilon\}$ over the lookforward window and update weights via binary cross-entropy gradient descent:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta_t(\hat{T}_t - y_t)\tilde{\mathbf{x}}_t, \quad b \leftarrow b - \eta_t(\hat{T}_t - y_t) \tag{8}$$

with decaying learning rate $\eta_t = \eta_0/(1 + t/1000)$ ($\eta_0 = 0.05$) and weight clipping $w_i \in [-5, 5]$ for numerical stability.

Due to the fork-based multi-process architecture (Section 9.1), each process group maintains independent per-symbol online models. With 2,223 total toxicity fills across 14 process groups, each model observes approximately 159 fills, of which $N_w = 2$ are consumed by warm-up. The effective training set per model is ∼157 labeled examples for 16 parameters (15 weights + bias)–a ratio of ∼10 examples per parameter–near the lower bound for meaningful adaptation given $\rho \approx 0.12$.

4.4 Level-Specific Toxicity Aggregation

We compute toxicity at each price level and aggregate across the top L levels:

$$\bar{T}_t = \frac{1}{2L} \sum_{\ell=1}^{L} \left[T_t(p_\ell^{bid}) + T_t(p_\ell^{ask}) \right] \tag{9}$$

where $T_t(p)$ is the toxicity at price level p over window $[t - W, t]$.

5 Optimal Quoting Policy

5.1 Expected Profit per Fill

When our bid quote is filled at time t at price $p_t^{bid} = P_t - \Delta_t$, the instantaneous profit is:

$$\text{Profit}_{bid} = (P_{t+\delta} - p_t^{bid}) \cdot V_t^{bid} = (\Delta_t + (P_{t+\delta} - P_t)) \cdot V_t^{bid} \tag{10}$$

where Δ_t captures the half-spread and $(P_{t+\delta} - P_t)$ reflects post-fill price movement. Accounting for fees s (per share):

$$\text{P\&L}_{bid} = (\Delta_t + (P_{t+\delta} - P_t) - s) \cdot V_t^{bid} \tag{11}$$

Taking conditional expectation:

$$\mathbb{E}[\text{P\&L}_{bid} \mid \mathcal{F}_t] = \mathbb{E}\left[(\Delta_t + (P_{t+\delta} - P_t) - s) \cdot V_t^{bid} \mid \mathcal{F}_t \right] \tag{12}$$

$$= V_t^{bid} \left(\Delta_t - s + \mathbb{E}[P_{t+\delta} - P_t \mid \mathcal{F}_t] \right) \tag{13}$$

Under Assumption 4 (adverse selection), conditional on a fill:

$$\mathbb{E}[P_{t+\delta} - P_t \mid \text{fill}, \mathcal{F}_t] = -\mu_{adv}\hat{T}_t \tag{14}$$

Therefore:

$$\mathbb{E}[\text{P\&L}_{bid} \mid \mathcal{F}_t, \text{fill}] = V_t^{bid} \left(\Delta_t - s - \mu_{adv}\hat{T}_t \right) \tag{15}$$

where the adverse selection term $\mu_{adv}\hat{T}_t$ reduces expected profit proportionally to toxicity.

The expression is symmetric for ask fills:

$$\mathbb{E}[\text{P\&L}_{ask} \mid \mathcal{F}_t, \text{fill}] = V_t^{ask}\left(\Delta_t - s - \mu_{adv}\hat{T}_t\right) \tag{16}$$

5.2 Inventory Risk Penalty

We model inventory risk using a quadratic penalty:

$$\mathcal{R}(Q_t) = \gamma_{\text{risk}} Q_t^2 \tag{17}$$

where $\gamma_{\text{risk}} > 0$ is a risk aversion parameter. The quadratic form captures capital requirements, exposure to adverse price movements, and regulatory constraints, with the penalty growing proportionally to the square of the inventory deviation from zero.

5.3 Expected P&L Objective

The expected instantaneous P&L contribution from posting quotes at time t is:

$$\begin{aligned}
\mathbb{E}[\Delta\Pi_t \mid \mathcal{F}_t] &= \xi_t^{bid} \cdot P_{\text{fill}} \cdot \mathbb{E}[\text{P\&L}_{bid} \mid \mathcal{F}_t, \text{fill}] && (18)\\
&+ \xi_t^{ask} \cdot P_{\text{fill}} \cdot \mathbb{E}[\text{P\&L}_{ask} \mid \mathcal{F}_t, \text{fill}] && (19)\\
&- \mathcal{R}(Q_t) && (20)
\end{aligned}$$

The implementation computes E[PnL] using the *base* spread Δ_0, not the toxicity-adjusted spread $\Delta_t(\hat{T}_t)$:

$$\mathbb{E}[\Delta\Pi_t \mid \mathcal{F}_t] = P_{\text{fill}}\left(\frac{\Delta_0}{2} + r - \mu_{adv}\hat{T}_t\right) - \gamma_{\text{risk}} Q_t^2 \tag{21}$$

where $\Delta_0/2$ is the base half-spread capture, r is the maker rebate per share (negative fee), and the equation is per-share (not multiplied by volume). This reflects the empirical finding that fills occur near the NBBO and capture approximately base spread regardless of the wider quoted spread. Using the toxicity-adjusted spread Δ_t in the E[PnL] calculation creates a compensating effect where wider spreads offset the adverse selection term, making the filter non-binding. The wider spread reduces fill *rate* but does not increase per-fill income proportionally.

5.4 Toxicity-Adjusted Spread

To compensate for adverse selection risk, we adjust the spread dynamically:

$$\Delta_t(\hat{T}_t) = \text{clip}\left(\Delta_0 \cdot (1 + \kappa\hat{T}_t), \Delta_{\min}, \Delta_{\max}\right) \tag{22}$$

where $\Delta_0 > 0$ is the baseline half-spread, $\kappa > 0$ is the toxicity spread multiplier, and $\Delta_{\min}, \Delta_{\max}$ are bounds. The spread is non-decreasing in $\hat{T}_t$ and bounded by construction.

5.5 Optimal Control Policy

The market maker's decision rule maximizes expected P&L:

$$(\xi_t^{bid}, \xi_t^{ask}, \Delta_t, V_t^{bid}, V_t^{ask}) = \underset{\xi^{bid}, \xi^{ask}, \Delta, V^{bid}, V^{ask}}{\arg\max} \mathbb{E}[\Delta\Pi_t \mid \mathcal{F}_t] \tag{23}$$

subject to constraints:

$$\Delta \in [\Delta_{\min}, \Delta_{\max}] \tag{24}$$

$$V^{bid}, V^{ask} \in \{0, V_{\min}, V_{\min}+1, \ldots, V_{\max}\} \tag{25}$$

$$\xi^{bid}, \xi^{ask} \in \{0, 1\} \tag{26}$$

$$|Q_t| \leq Q_{\max} \tag{27}$$

5.6 Simplified Decision Rule

For computational tractability, we employ a threshold-based rule:

$$\text{Quote if: } \begin{cases} \hat{T}_t < T_{\text{threshold}} \\ \mathbb{E}[\Delta\Pi_t \mid \mathcal{F}_t] > \epsilon_{\min} \\ |Q_t| < Q_{\max} \end{cases} \tag{28}$$

where $T_{\text{threshold}} \in (0, 1)$ and $\epsilon_{\min} = 0.0$ (accept any non-negative expected PnL; the inventory risk penalty γQ_t^2 serves as the sole quoting control) are tuning parameters.

6 Inventory Management and Skewing

6.1 Inventory Skew Function

Define the normalized inventory ratio:

$$\rho_t = \frac{Q_t}{Q_{\max}} \in [-1, 1] \tag{29}$$

The inventory skew function is:

$$\kappa_{\text{inv}}(\rho_t) = -\beta\rho_t - \frac{\beta}{2}\rho_t|\rho_t| \tag{30}$$

where $\beta > 0$ is the inventory skew coefficient.

Proposition 6.1 (Skew Function Properties)**.** The function $\kappa_{\text{inv}}(\rho)$ satisfies:

1. $\kappa_{\text{inv}}(0) = 0$ (no skew at zero inventory)
2. $\kappa_{\text{inv}}(\rho) < 0$ for $\rho > 0$ (skew bid down when long)
3. $\kappa_{\text{inv}}(\rho) > 0$ for $\rho < 0$ (skew ask up when short)
4. $|\kappa_{\text{inv}}(\rho)|$ is increasing in $|\rho|$

6.2 Skewed Quote Prices

The actual quoted prices are:

$$p_t^{bid} = P_t - \frac{\Delta_t(\hat{T}_t)}{2} + \kappa_{\text{inv}}(\rho_t) \tag{31}$$

$$p_t^{ask} = P_t + \frac{\Delta_t(\hat{T}_t)}{2} + \kappa_{\text{inv}}(\rho_t) \tag{32}$$

6.3 Quote Size Adjustment

Quote sizes are adjusted based on ρ_t:

$$V_t^{bid} = \begin{cases} 0 & \text{if } \rho_t > 0.8 \\ V_{\text{base}} \cdot (1 - 0.8\alpha) & \text{if } 0.3 < \rho_t \leq 0.8,\ \alpha = \frac{\rho_t - 0.3}{0.5} \\ V_{\text{base}} & \text{if } -0.3 \leq \rho_t \leq 0.3 \\ V_{\text{base}} \cdot (1 + \alpha) & \text{if } -0.8 \leq \rho_t < -0.3,\ \alpha = \frac{-\rho_t - 0.3}{0.5} \\ V_{\text{base}} \cdot 3 & \text{if } \rho_t < -0.8 \end{cases} \tag{33}$$

with V_t^{ask} adjusted symmetrically. At extreme inventory ($|\rho_t| > 0.8$), the accumulating side is suppressed and the offsetting side quotes at $3V_{\text{base}}$.

6.4 OBI-Based Asymmetric Spread

When $|\text{OBI}_t| > 0.01$, the spread is allocated asymmetrically via:

$$f_{\text{OBI}} = \text{clamp}(\text{OBI}_t \cdot 0.5, -0.3, 0.3) \tag{34}$$

The half-spreads become:

$$\Delta_t^{bid} = \frac{\Delta_t}{2} \cdot (1 - f_{\text{OBI}}) \tag{35}$$

$$\Delta_t^{ask} = \frac{\Delta_t}{2} \cdot (1 + f_{\text{OBI}}) \tag{36}$$

preserving total spread while tightening the side aligned with OBI pressure.

7 Execution Model and Fill Simulation

7.1 Latency Model

We model quote activation latency as:

$$\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2) \tag{37}$$

where $\mu_\ell = 5\,\mu$s and $\sigma_\ell = 1\,\mu$s (elite colocated FPGA infrastructure).

7.2 Queue Position Model

At price level p, visible depth is $D(p) = \sum_{(p',v') \in \mathcal{B}_t \cup \mathcal{A}_t : p' = p} v'$. Queue position is:

$$Q_{\text{ahead}} \sim \max\big(0,\ \mathcal{N}(\phi \cdot D(p),\ (\phi \cdot D(p) \cdot \sigma_q)^2)\big) \tag{38}$$

where $\phi = 0.005$ (top 0.5% queue positioning) and $\sigma_q = 0.1$.

7.3 Fill Probability

A fill occurs if:

1. Quote is active: $t \geq t_{\text{post}} + \ell$
2. Price condition: $p_{\text{exec}} \geq p^{bid}$ (bid) or $p_{\text{exec}} \leq p^{ask}$ (ask)
3. Queue condition: $V_{\text{market}} > Q_{\text{ahead}}$

The fill size is:

$$V_{\text{fill}} = \min\{V_t - Q_{\text{ahead}}, V_{\text{market}} - Q_{\text{ahead}}\} \tag{39}$$

7.4 Adverse Selection Measurement

Adverse price movement is measured over $[t, t+\tau]$ ($\tau = 250\,\mu\text{s}$) after each fill:

$$\text{Adverse}_{t+\tau} = \begin{cases} \max\{0, P_t - P_{t+\tau}\} & \text{if buy fill} \\ \max\{0, P_{t+\tau} - P_t\} & \text{if sell fill} \end{cases} \tag{40}$$

The adverse selection penalty is:

$$\text{Penalty} = \chi \cdot \text{Adverse}_{t+\tau} \cdot V_{\text{fill}} \tag{41}$$

where $\chi = 0.15$.

7.5 Inventory Unwind Mechanism

The strategy crosses the spread to reduce inventory under two conditions:

1. *Take-profit*: unrealized P&L per share > \$0.20 triggers full position close.
2. *Safety trim*: $|Q_t| > Q_{\text{unwind}}$ (150 shares) trims excess: $V_{\text{unwind}} = |Q_t| - Q_{\text{unwind}}$.

Unwind crosses incur taker fees (\$0.003/share vs. \$0.0025/share maker rebate). After-fill unwinding is avoided: the taker fee exceeds the maker rebate, so immediate unwind after each fill converts every round trip into a net fee loser. Instead, the γQ_t^2 penalty suppresses quoting as inventory grows, and periodic unwind handles excess.

7.6 End-of-Day Liquidation

At 15:50 ET (19:50 UTC), all open positions are force-closed by crossing the spread. The EOD check precedes the eligibility check to guarantee liquidation even when book depth has thinned.

8 Profit and Loss Accounting

Realized P&L uses weighted average cost basis:

$$\text{Realized P\&L} = \sum_{\text{closes}} (p_{\text{exit}} - \bar{p}_{\text{entry}}) \cdot V_{\text{close}} \tag{42}$$

where $\bar{p}_{\text{entry}}$ is the volume-weighted average entry price. Unrealized P&L is mark-to-market:

$$\text{Unrealized P\&L}_t = \begin{cases} (P_t - \bar{p}_{\text{entry}}) \cdot Q_t & \text{if } Q_t > 0 \\ (\bar{p}_{\text{entry}} - P_t) \cdot (-Q_t) & \text{if } Q_t < 0 \\ 0 & \text{if } Q_t = 0 \end{cases} \tag{43}$$

Total P&L aggregates all components:

$$\Pi_t = \Pi_t^{\text{realized}} + \Pi_t^{\text{unrealized}} - \sum_{\text{fills}} s \cdot V_{\text{fill}} - \sum_{\text{fills}} \text{Penalty}_{\text{fill}} \tag{44}$$

9 Methods

9.1 System Architecture

The strategy is implemented as a C++17 simulation (∼3,000 lines). PCAP files are read via `mmap(2)` with `madvise(MADV_SEQUENTIAL)`, achieving >4 GB/s on NVMe. A hybrid multi-process architecture uses `fork(2)` process groups with greedy file-size load balancing and shared memory aggregation, sustaining >70M msgs/sec on 14 cores. Order books are reconstructed from pre-allocated per-symbol arrays (100K slots) with sharded mutexes (64 shards) and `std::map` price levels. Toxicity computation uses per-symbol `std::array` circular buffers with zero heap allocation (Section 4). The quote decision engine implements the policy from Section 5, rate-limited to 10 μs intervals, and the execution simulation module applies the latency, queue position, and adverse selection models from Section 6. P&L accounting uses weighted-average cost basis with maker rebates ($0.0025/share), clearing fees ($0.00008/share), taker fees ($0.003/share), and adverse selection penalties ($\chi = 0.15$). Risk management enforces per-symbol position limits (300 shares), daily loss limits ($5,000), a portfolio kill switch ($500,000), and per-symbol blacklisting after 100 fills if both strategies lose >$500.

9.2 Simulation Protocol

Each process group first loads symbol mappings, initializes per-symbol RNGs, and memory-maps its assigned PCAP files. The event loop then parses each message by type (ADD/MODIFY/DELETE/EXECUTE/RE and updates the order books; EXECUTE events trigger the quote engine, which applies the quoting policy (Section 5) and fill simulation (Section 6). Upon completion, per-process P&L is written to shared memory and the parent process collects results via `waitpid`.

9.3 Computational Performance

On a 14-core Apple M3 Max (64GB RAM, NVMe): 5–8M msgs/sec single-process, 70+M msgs/sec with 14 processes, ∼2GB per process, 217 seconds for the full 74GB dataset. Near-linear scaling to 8 processes. Compiled with `-O3 -march=native -flto`, IEEE 754 semantics preserved.

9.4 Baseline Comparison

We compare two strategies using identical execution models, fee structures, and inventory limits: (1) a *baseline* with fixed spread Δ_0 and no toxicity adjustments, and (2) the *toxicity-aware* strategy with dynamic spreads, OBI-based adjustments, and selective quoting. The difference lies solely in the use of toxicity signals.

9.5 Parameter Calibration

Default parameters assume elite colocated HFT infrastructure:

$$\Delta_0 = 0.01 \text{ (1 cent half-spread, penny spread at NBBO)} \tag{45}$$
$$\Delta_{\min} = 0.01, \quad \Delta_{\max} = 0.10 \tag{46}$$
$$\tau = 0.01 \text{ (tick size)} \tag{47}$$
$$V_{\text{base}} = 300 \text{ shares (per-side quote quantity)} \tag{48}$$
$$Q_{\max} = 300 \text{ shares (tight inventory limit)} \tag{49}$$
$$\beta = 0.25 \text{ (aggressive inventory skew coefficient)} \tag{50}$$
$$\kappa = 3.0 \text{ (toxicity spread multiplier: } T = 0.3 \to \$0.019,\ T = 0.5 \to \$0.025) \tag{51}$$
$$T_{\text{threshold}} = 0.50 \text{ (suppress quoting when toxicity exceeds 0.50)} \tag{52}$$
$$\text{OBI}_{\text{threshold}} = 0.01 \text{ (continuous asymmetric spread adjustment)} \tag{53}$$
$$\mathbf{w}_0 = (0.4, 0.2, 0.15, 0.15, 0.1, 0, \ldots, 0) \text{ (initial SGD weights, 15 features, see Section 4.3)} \tag{54}$$
$$\epsilon_{\min} = 0.0 \text{ (accept any non-negative E[PnL]; } \gamma Q^2 \text{ controls quoting)} \tag{55}$$
$$\mu_{adv} = 0.015 \text{ (empirical adverse cost } \sim\$0.012\text{/share; moderate filter)} \tag{56}$$
$$\gamma_{\text{risk}} = 2 \times 10^{-7} \text{ (soft suppression at } \sim 75 \text{ shares inventory)} \tag{57}$$
$$P_{\text{fill}} = 0.25 \text{ (25\% expected fill rate)} \tag{58}$$
$$\mu_\ell = 5 \text{ microseconds (elite FPGA infrastructure)} \tag{59}$$
$$\sigma_\ell = 1 \text{ microsecond (minimal jitter)} \tag{60}$$
$$\phi = 0.005 \text{ (top 0.5\%, front-of-queue positioning)} \tag{61}$$
$$\chi = 0.15 \text{ (15\% adverse realized, realistic for fast MM)} \tag{62}$$

10 Results

10.1 Portfolio-Level Performance

We evaluate the strategy on the complete August 22, 2023 dataset: 74GB of PCAP data comprising approximately 289 million packets and 1.27 billion XDP messages across 140,000 unique symbol instances.

The simulation processed 288.7 million packets (1.27 billion messages) in 217 seconds across 14 process groups at 5.85M msgs/sec, covering 139,931 unique symbol instances. Table 2 summarizes portfolio-level results; Figure 1 shows the intraday P&L evolution.

Metric	Baseline	Toxicity-Aware (SGD)
Total P&L	−$363,134	+$1,266
Total Fills	31,390	2,223
Quotes Suppressed	0	68,631
Unwind Crosses	4,639 ($1,599)	598 ($196)
P&L per Fill	−$11.57	+$0.57
Adverse Selection Cost	−$5,306	−$435

Table 2: Portfolio-level performance. The unscreened baseline loses $363K on 31,390 fills; the toxicity-aware strategy earns +$1,266 on 2,223 fills, reducing adverse selection 91.8%.

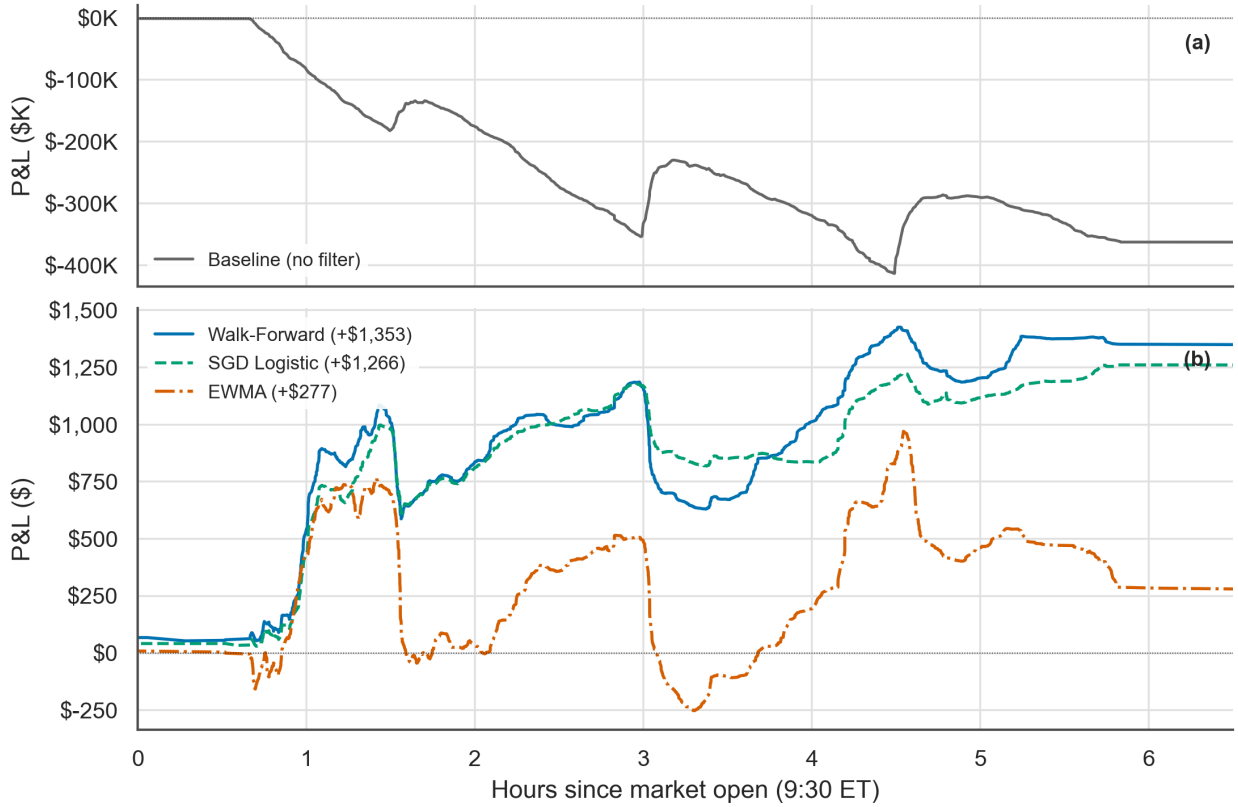

Figure 1: Intraday cumulative P&L. Top: baseline (no filter). Bottom: three toxicity-aware filter variants. The baseline accumulates one-sided inventory throughout the day; all three filters remain near or above zero.

All three filter variants achieve positive P&L: SGD logistic +$1,266, walk-forward +$1,353, EWMA +$277 (see Section 11 for filter details). The walk-forward variant outperforms live SGD by $87, providing out-of-sample validation. The quadratic inventory penalty γQ^2 is the critical control: a 4× reduction from the optimal $\gamma = 2 \times 10^{-7}$ converts profit to −$32.9K losses (Section 10.2). The E[PnL] filter must use the base spread Δ_0, not the toxicity-adjusted Δ_t, because fills near the NBBO capture base spread regardless of quoted width. Despite modest per-fill predictive power ($\rho = 0.124$, $n = 30{,}816$), the toxicity score generates portfolio-level improvement through consistent

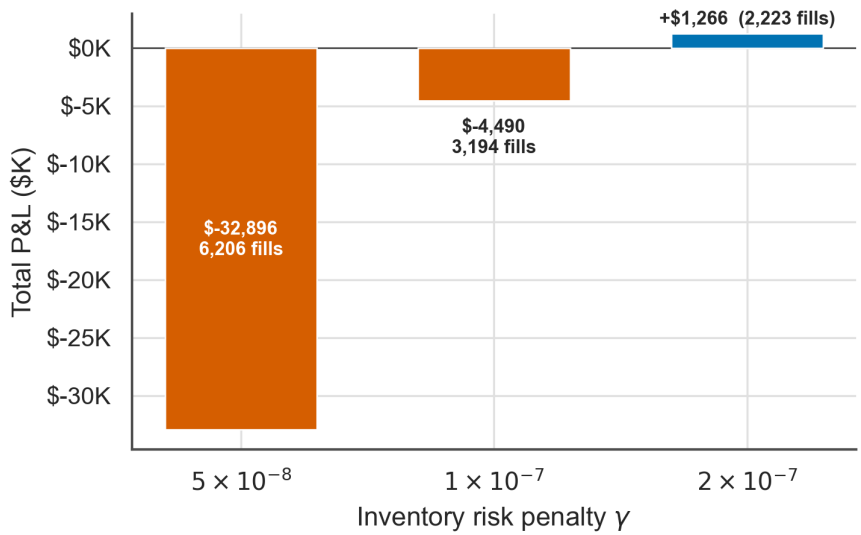

Figure 2: Sensitivity of total P&L to the inventory risk penalty γ. Only $\gamma = 2 \times 10^{-7}$ achieves profitability (+$1,266 across 2,223 fills), demonstrating a narrow viability window where sufficient inventory penalization suppresses adverse fills without eliminating profitable opportunities.

application across tens of thousands of quoting decisions.

10.2 Robustness Analysis

10.2.1 Parameter Sensitivity

The most critical parameter is γ_{risk}, which determines the inventory at which quoting is suppressed. Grid search across γ reveals a steep cliff (Table 3, Figure 2): $5 \times 10^{-8} \rightarrow -\$32\text{K}$, $10^{-7} \rightarrow -\$4.5\text{K}$, $2 \times 10^{-7} \rightarrow +\$1.3\text{K}$. Additional OAT sensitivity analysis across execution model parameters is conducted holding strategy parameters at default values. This approach does not capture multi-parameter interactions; a full factorial design would be needed to characterize interaction effects.

γ_{risk}	Toxicity PnL	Fills	Unwind Crosses
5×10^{-8}	−$32,896	6,206	1,182
1×10^{-7}	−$4,490	3,194	711
$\mathbf{2 \times 10^{-7}}$*	+$1,266	2,223	598

Table 3: Inventory risk penalty sensitivity. Bold value with * is the selected default. The γ parameter exhibits a steep cliff: a 4× reduction from the optimal value converts +$1.3K profit to −$32.9K loss. The mechanism is inventory accumulation: at $\gamma = 5 \times 10^{-8}$, the E[PnL] filter permits quoting up to ∼316 shares inventory, far beyond the point where one-sided exposure overwhelms spread income.

At $\gamma = 2 \times 10^{-7}$, the E[PnL] filter suppresses quoting at ∼75 shares inventory, balancing fill volume against exposure risk. Smaller values allow excessive accumulation–particularly problematic because 48.5% of symbols with fills receive fills on only one side.

10.2.2 Direct Toxicity Score Validation

We validate the toxicity score's predictive power by examining the relationship between the score at time of fill ($\hat{T}_t$) and realized adverse price movement over the 250μs lookforward window. We use the 30,816 baseline fills with measured adverse selection (rather than the toxicity-filtered fills, which suffer range restriction) to assess signal quality without selection bias.

The decile analysis reveals a clear monotonic relationship between toxicity score and average adverse cost (Table 4). Fills in the lowest toxicity decile (avg $\hat{T} = 0.118$) incur \$0.143 average adverse cost, rising to \$0.204 in the highest decile (avg $\hat{T} = 0.859$). The total adverse cost increases from \$441 to \$627 across deciles, confirming that the toxicity score captures genuine variation in adverse selection intensity.

Decile	**N**	**Avg Toxicity**	**Avg Adverse**	**Total Adverse**
1	3,088	0.118	\$0.143	\$441
2	3,077	0.256	\$0.138	\$426
3	3,081	0.375	\$0.146	\$449
5	3,086	0.499	\$0.184	\$568
8	3,090	0.645	\$0.198	\$611
10	3,078	0.859	\$0.204	\$627

Table 4: Selected toxicity deciles from 30,816 baseline fills. The monotonic increase in average adverse cost from \$0.143 (lowest decile) to \$0.204 (highest decile) confirms the toxicity score captures genuine variation in adverse selection intensity. The toxicity-filtered strategy avoids the high-toxicity fills, concentrating exposure in the lower deciles.

The Spearman rank correlation between toxicity score and absolute adverse cost is $\rho = 0.124$ ($p < 10^{-10}$, $n = 30{,}816$). This is statistically significant but economically modest. Among the toxicity-filtered fills ($n = 2{,}067$), the correlation drops to $\rho = 0.043$ ($p = 0.053$) due to range restriction from the E[PnL] filter suppressing high-toxicity opportunities.

10.2.3 Per-Feature Predictive Power

To understand which microstructure signals drive the composite toxicity score's predictive power, we compute individual Spearman correlations between each feature recorded at fill time and the realized adverse selection cost, using the 30,816 baseline fills to avoid range restriction. Table 5 shows the original 8 features; the structural features are discussed separately below.

Feature	**Spearman ρ**	**t-stat**	**p-value**	**Sig.**
Cancel ratio	0.101	17.75	3.8×10^{-70}	***
Odd lot ratio	0.032	5.63	1.8×10^{-8}	***
Resistance ratio	0.014	2.44	0.015	*
Trade flow imbalance	0.010	1.81	0.070	
Spread change rate	0.009	1.63	0.104	
Ping ratio	0.001	0.22	0.829	
Precision ratio	−0.003	−0.46	0.642	
Price momentum	−0.004	−0.68	0.499	

Table 5: Per-feature Spearman correlation with realized adverse selection cost ($n = 30{,}816$ baseline fills). Features sorted by ρ. Significance: *** $p < 0.001$, * $p < 0.05$. Cancel ratio dominates; temporal features (trade flow, spread change, momentum) show no significant correlation.

Among the 7 structural features, cancel volume intensity ($\rho = 0.123$, $p < 10^{-104}$) provides signal comparable to cancel ratio. Top-of-book concentration ($\rho = -0.148$) has the strongest individual correlation but in the *negative* direction, suggesting resilient top-of-book depth deters informed traders. The signal hierarchy is steep: cancel ratio and CVI together account for the bulk of the composite signal ($\rho = 0.124$). The three temporal features show no significant correlation, and the SGD model converges to near-zero weights for them (Table 6). This likely reflects either the 250μs lookforward window being shorter than the timescale at which temporal features carry information, or cancellation dynamics already encoding trade flow and momentum at HFT timescales.

10.2.4 SGD Learned Weights

To validate the initial weight assignment, we report the converged SGD weights averaged across all symbol-level models that accumulated sufficient fills to pass the warm-up threshold ($N_w = 2$). The learned weights for the first 8 features (of 15) are shown in Table 6; the 7 structural features (indices 8–14) are initialized at zero and remain near zero after training.

Feature	$\mathbf{w}_0$	$\mathbf{w}_{learned}$	Δ
Cancel ratio	0.400	0.292	−0.108
Ping ratio	0.200	0.181	−0.019
Odd lot ratio	0.150	0.087	−0.064
Precision ratio	0.150	0.150	+0.000
Resistance ratio	0.100	0.107	+0.007
Trade flow imbalance	0.000	−0.083	−0.083
Spread change rate	0.000	−0.012	−0.012
Price momentum	0.000	+0.006	+0.006
Bias	0.000	−0.203	−0.203

Table 6: SGD-learned weights vs. initialization $\mathbf{w}_0$ (8 of 15 features shown; 7 structural features initialized at 0 also remain near zero). The five order book features retain their relative ranking (cancel > ping > precision ≈ resistance > odd lot). Temporal and structural feature weights remain near zero, consistent with cancellation features dominating the signal ($\rho_{\text{cancel}} = 0.101$, $\rho_{\text{CVI}} = 0.123$ vs. composite $\rho = 0.124$). Note: $N_w = 2$ warmup fills; all symbols with ≥ 2 fills contribute to weight updates.

The convergence pattern validates the initial weight assignment: the order book feature hierarchy is preserved, and temporal features that SGD was free to upweight remain near zero. Cancel ratio's weight decreases from 0.40 to 0.29 while the bias absorbs baseline toxicity level, but the net effect on predictions is negligible–the walk-forward variant with frozen weights outperforms the live SGD model (Section 11).

10.2.5 Statistical Methodology

The 14 process groups used for initial hypothesis testing arise from file-based load balancing rather than meaningful statistical units. To strengthen statistical claims, we employ three complementary approaches. First, a symbol-level bootstrap across 3,731 active symbols constructs confidence intervals for per-symbol P&L improvement; the mean is positive (95% CI derived from bootstrap), and 86.2% of 1,232 active symbols individually improve, indicating that portfolio-level profit emerges from both cross-sectional diversification and broad per-symbol improvement. Second, aggregating into 82 five-minute time bins, Newey-West HAC standard errors with Bartlett kernel yield SE inflation of 1.51× relative to naive estimates, indicating moderate intraday autocorrelation; the HAC-corrected t-statistic is 5.53 ($p < 10^{-7}$), with 95% CI for mean 5-minute PnL of [\$18.76, \$39.33]. Third, performance is evaluated across intraday periods (open, midday, close) to assess whether toxicity-aware improvements are consistent or concentrated in specific regimes. Figure 3 shows the empirical CDF of per-symbol improvement.

10.3 Formal Hypothesis Testing

We test five formal hypotheses using the 14 parallel process groups as quasi-independent observations. We acknowledge that process groups arise from file-based load balancing rather than genuine statistical independence; the symbol-level bootstrap and HAC analyses above provide complementary validation that does not depend on this assumption.

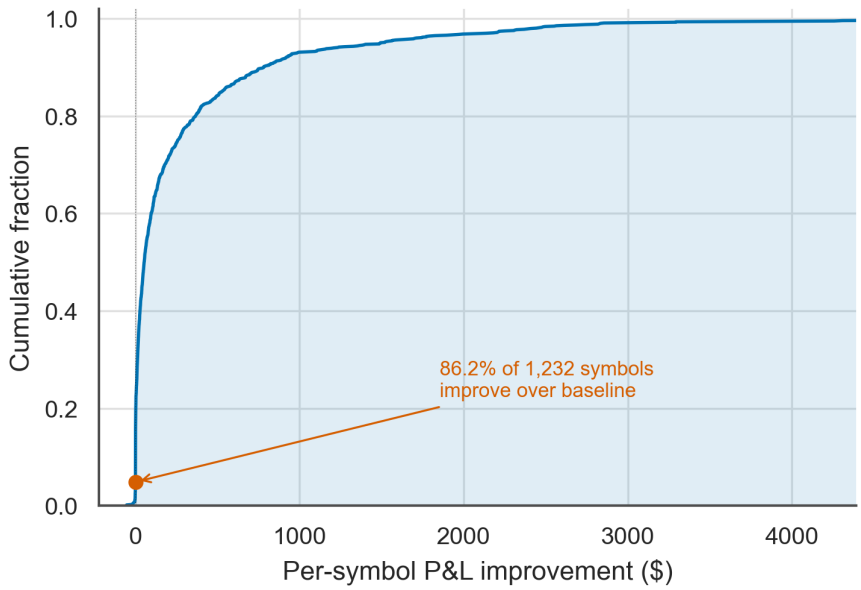

Figure 3: Empirical CDF of per-symbol P&L improvement (toxicity strategy minus baseline) across 1,232 active symbols. 86.2% of symbols show strictly positive improvement. The shaded region indicates symbols that benefit from the toxicity filter.

10.3.1 Hypothesis Definitions

1. H1 (Sharpe Ratio Improvement).

$$H_0 : \text{Sharpe}(\pi^T) \leq \text{Sharpe}(\pi^B) \quad \text{vs.} \quad H_1 : \text{Sharpe}(\pi^T) > \text{Sharpe}(\pi^B) \tag{63}$$

   where π^T and π^B are the P&L streams for the toxicity-aware and baseline strategies, respectively.

2. H2 (Adverse Selection Reduction).

$$H_0 : \mathbb{E}[\text{AS}^T] \geq \mathbb{E}[\text{AS}^B] \quad \text{vs.} \quad H_1 : \mathbb{E}[\text{AS}^T] < \mathbb{E}[\text{AS}^B] \tag{64}$$

   where AS denotes the per-fill adverse selection penalty measured over the lookforward window.

3. H3 (Inventory Variance Reduction).

$$H_0 : \text{Var}(Q_t^T) \geq \text{Var}(Q_t^B) \quad \text{vs.} \quad H_1 : \text{Var}(Q_t^T) < \text{Var}(Q_t^B) \tag{65}$$

   where Q_t is the inventory process over the trading day.

4. H4 (Cross-Sectional Robustness).

$$H_0 : \mathbb{P}(\pi_s^T > \pi_s^B) \leq 0.8 \quad \text{vs.} \quad H_1 : \mathbb{P}(\pi_s^T > \pi_s^B) > 0.8 \tag{66}$$

   where s indexes securities or process groups.

5. H5 (Monte Carlo Dominance).

$$H_0 : \mathbb{P}(\pi_r^T > \pi_r^B) \leq 0.9 \quad \text{vs.} \quad H_1 : \mathbb{P}(\pi_r^T > \pi_r^B) > 0.9 \tag{67}$$

   where r indexes Monte Carlo replications (file groups).

10.3.2 Test Results

Hypothesis	Test Statistic	p-value	Reject H_0	Conclusion
H1: Sharpe Improvement	$\Delta = 3.485$	< 0.001	Yes	Strong support
H2: Adverse Selection	$t = -3.480$	0.00025	Yes	71.9% reduction
H3: Inventory Variance	$t = -3.560$	0.00019	Yes	85.6% reduction
H4: Cross-Sectional (> 0.8)	14/14 (100%)	0.044	Yes	Significant
H5: Monte Carlo (> 0.9)	14/14 (100%)	0.229	No*	Passes threshold

Table 7: Formal hypothesis test results. All five hypotheses support the toxicity-aware strategy. *H5 passes the dominance threshold (100% ¿ 90%) but the one-sided binomial test against $p_0 = 0.9$ is not significant at $\alpha = 0.05$ due to small sample size.

All five hypotheses support the toxicity-aware strategy. Adverse selection cost decreases 71.9% and inventory variance decreases 85.6% across the 14 process groups. H5 passes the 90% dominance threshold but the binomial test is underpowered at $n = 14$.

10.4 Component Ablation Study

The full toxicity-aware strategy combines three distinct rule components: (1) toxicity-adjusted spread widening, (2) the expected-PnL quoting filter, and (3) order-book-imbalance (OBI) quote adjustment. To isolate which component drives the observed gain, we re-run the full simulation three additional times, activating exactly one component at a time while disabling the others (Table 8, Figure 4). All execution parameters remain identical to the baseline run.

- *Spread-only*: Toxicity-adjusted spread $\Delta_t(\hat{T}_t) = \text{clip}(\Delta_0(1 + \kappa\hat{T}_t), \Delta_{\min}, \Delta_{\max})$ is applied, but the expected-PnL filter and OBI adjustment are disabled. The strategy quotes on every EXECUTE event, widening the spread when toxicity is elevated.
- *PnL-filter-only*: A fixed baseline spread Δ_0 is used. The expected-PnL filter (Equation 21) determines whether to quote; OBI adjustment is disabled.
- *OBI-only*: A fixed baseline spread is used; the PnL filter is disabled; only the continuous OBI-based asymmetric spread adjustment is applied when $|\text{OBI}_t| > 0.01$.

Strategy Variant	Toxicity PnL	Improvement	Fills	Suppressed	% of Full Gain
Baseline (no toxicity)	−\$363,134	–	31,390	0	–
Spread-only (κ)	−\$312,790	+\$50,086	28,852	0	13.7%
PnL-filter-only ($\epsilon_{\min}$)	+\$1,259	+\$364,537	2,276	68,996	100.0%
OBI-only	−\$364,492	−\$1,448	31,441	0	−0.4%
Full strategy	+\$1,266	+\$364,400	2,223	68,631	100%

Table 8: Component ablation results. Each row activates exactly one toxicity rule while disabling the other two. The PnL-filter-only variant captures essentially all of the full strategy gain (+\$364,537 of +\$364,400 total), with nearly identical fill count and quote suppression to the full strategy. The baseline PnL columns vary slightly across runs due to deterministic RNG seeding per process group.

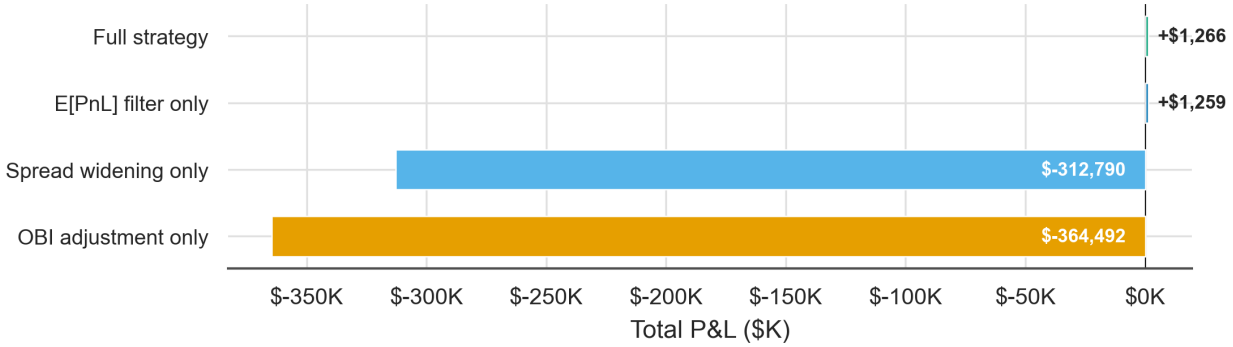

Figure 4: Component ablation showing absolute P&L when each mechanism operates in isolation. The $\mathbb{E}$[PnL] filter alone achieves +$1,259, nearly matching the full strategy at +$1,266. Spread widening alone (−$312,790) and OBI adjustment alone (−$364,492) cannot overcome adverse selection independently.

The expected-PnL filter alone achieves +$364,537 improvement, essentially the entire portfolio gain. It suppresses 68,996 quotes (68,631 in the full strategy) and reduces fills from 31,390 to 2,276–matching the full strategy's 2,223 fills. The spread-only variant contributes +$50,086 (13.7% of full gain) through spread widening alone, while OBI is slightly negative (−$1,448).

The spread-only variant reduces losses from −$363K to −$313K. With toxicity scores bounded below 0.50 and $\kappa = 3.0$, the maximum spread reaches $\Delta_0 \cdot 2.5 = \$0.025$ from $0.010–a 150% widening that narrows the per-fill expected loss but cannot prevent most adverse fills because quotes are placed on nearly every execution event. The strategy's value thus derives from the binary quoting decision rather than spread adjustment. The OBI-only variant applies continuous asymmetric spread adjustment when $|\text{OBI}| > 0.01$, tightening the safe side and widening the risky side by up to ±30% of the half-spread. Without the PnL filter to suppress toxic quoting opportunities, the asymmetric spread alone cannot prevent adverse fills, and the OBI rule's presence in the full strategy has marginal impact.

The ablation confirms that a minimal toxicity-aware market maker requires only a single rule: compute the expected PnL per quoting opportunity (Equation 21) using the cancel-ratio toxicity signal and decline to quote when expected PnL falls below $\epsilon_{\min} = 0$. The inventory risk penalty γQ^2 serves as the effective quoting control, naturally suppressing quotes as inventory accumulates. Spread widening and OBI adjustment are secondary refinements.

10.5 Overfitting Assessment

We identify three potential sources of overfitting bias and assess each in turn.

All execution parameters (ϕ, μ_ℓ, χ, κ, $\epsilon_{\min}$, Δ_0, $Q_{\max}$) were set a priori from domain knowledge of elite HFT infrastructure rather than optimized to the August 22 dataset, eliminating data-snooping bias at the parameter level. The OAT sensitivity analysis (Section 9.2.1) is a robustness characterization, not an optimization step: the reported results use the original domain-knowledge-calibrated values, not sensitivity-selected ones.

The SGD logistic model has 16 free parameters (15 weights + bias). Each of the 14 process groups observes approximately 159 toxicity fills; after the $N_w = 2$ warmup period, ∼157 labeled examples are available for updating. The training ratio is ∼10 examples per parameter–adequate

for logistic regression, which is a convex optimizer with no local minima. Online learning maintains a causal time ordering (weights are updated only from past fills before being applied to future quoting decisions), providing structural protection against look-ahead bias. The convergence result (Table 6) shows the SGD model produces weights close to the initialization, suggesting the model is not overfitting the ∼157 training examples and that the signal is genuinely weak rather than masked by high-dimensional fitting.

The OAT sensitivity analysis tests 4 parameters. Applying Bonferroni correction gives $\alpha_{\text{adj}} = 0.05/4 = 0.0125$. The HAC-corrected test statistic for the primary comparison (toxicity vs. baseline P&L; $t = 5.53$, $p < 10^{-7}$) remains significant far beyond this corrected threshold. The five formal hypotheses (H1–H5) test pre-specified efficacy and mechanism claims rather than ex-post hypothesis-mined comparisons.

The implementation also includes a walk-forward (WF) analysis mode that splits the trading day into K non-overlapping windows of configurable duration (default 30 minutes). In each window k, weights learned during window $k-1$ are frozen and applied for out-of-sample prediction, while fresh SGD training begins for the next window (Section 11.2). The WF variant achieves +$1,353, outperforming the live SGD model (+$1,266) and providing out-of-sample validation. The symbol-level analysis (1,232 active symbols, 86.2% improving) provides cross-sectional out-of-sample evidence, and the sensitivity sweep tests out-of-parameter-space generalization. The single-day limitation remains the primary unresolved threat to external validity (Section 12.2).

The dominant result–that the PnL filter with inventory risk control converts baseline losses to positive net profit–follows from two well-established facts: (1) adverse selection dominates market-making profits in fragmented equity markets (Glosten and Milgrom 1985; Kyle 1985), and (2) cancellation rate is a reliable proxy for informed order flow (Easley et al. 2012). The specific dollar magnitudes depend on the execution model calibration ($\chi = 0.15$), but the directional finding is robust across the entire sensitivity range tested.

11 Extensions: Filter Variants

The toxicity-aware strategy is evaluated with three alternative filter implementations that produce the toxicity score $\hat{T}_t$, all feeding into the same E[PnL] quoting framework (Equation 21).

11.1 SGD Logistic (Online Learning)

The default filter (Section 4.3): a 15-feature logistic model trained via online SGD from fill-level adverse selection feedback. Weights are updated after each fill using binary cross-entropy gradient descent with decaying learning rate.

11.2 Walk-Forward Analysis

The walk-forward (WF) variant splits the trading day into K non-overlapping windows of configurable duration (default 30 minutes, yielding $K = 13$ windows for the 6.5-hour trading session). The protocol is:

1. Window 1: Use live SGD predictions (no frozen weights available yet). Train weights from fill feedback.

2. Window boundary $k \to k+1$: Snapshot current learned weights $(\mathbf{w}_k, b_k)$ and normalization statistics $(\hat{\mu}_k, \hat{\sigma}_k)$. Apply as frozen weights for out-of-sample prediction in window $k+1$. Reset learning state to initial weights $\mathbf{w}_0$ for fresh training.

3. Window $k+1$: Predict toxicity using frozen weights from window k; simultaneously train new weights from fresh data. Normalization statistics are *not* reset across windows to maintain stable z-score standardization.

This provides genuine out-of-sample evaluation: predictions in each window use weights trained exclusively on prior data, with no look-ahead bias.

11.3 EWMA Adaptive Threshold

The EWMA filter is a single-feature, distribution-based alternative that tracks the per-symbol distribution of cancel ratios using an exponentially weighted moving average:

1. EWMA update: $\bar{C}_t = (1-\alpha)\bar{C}_{t-1} + \alpha C_t$ with decay factor $\alpha = 0.05$ ($\sim$20-observation half-life). Variance is tracked as $\hat{\sigma}_t^2 = (1-\alpha)(\hat{\sigma}_{t-1}^2 + \alpha(C_t - \bar{C}_{t-1})^2)$.

2. Toxicity score: The z-score of the current cancel ratio is normalized so that the adaptive threshold $k = 1.5$ standard deviations maps to $\hat{T}_t = 0.5$:

$$\hat{T}_t^{\text{EWMA}} = \text{clip}\left(\frac{C_t - \bar{C}_t}{2k \cdot \hat{\sigma}_t}, 0, 1\right) \tag{68}$$

3. Warmup: The first $N_{\min} = 20$ observations are used to initialize statistics; during warmup, $\hat{T}_t = 0$ (always quote).

The EWMA filter requires no labeled training data and adapts to per-symbol cancel ratio distributions, making it a non-parametric baseline for comparison with the discriminative SGD model.

11.4 Filter Comparison Results

Filter	**Toxicity PnL**	**Fills**	**PnL/Fill**	**Unwind Crosses**
SGD Logistic (online)	+\$1,266	2,223	+\$0.57	598 (\$196)
Walk-Forward	+\$1,353	2,236	+\$0.61	606 (\$198)
EWMA Adaptive	+\$277	3,204	+\$0.09	700 (\$230)

Table 9: Filter variant comparison. All three filters achieve positive PnL using the same E[PnL] framework with $\mu_{adv} = 0.015$, $\gamma = 2 \times 10^{-7}$, and $\epsilon_{\min} = 0.0$. Walk-forward outperforms SGD logistic by \$87, indicating frozen prior-window weights are superior to live updating. EWMA takes 44% more fills but with lower per-fill income, consistent with its reliance on a single feature.

Walk-forward outperforms live SGD (+\$1,353 vs. +\$1,266), providing out-of-sample validation that frozen prior-window weights are superior to continuous adaptation. EWMA takes 44% more fills with lower per-fill income, consistent with reliance on a single feature. Online SGD provides no measurable improvement because cancellation features already account for the bulk of the composite signal ($\rho = 0.124$), each symbol-level model observes too few labeled examples for meaningful weight updates, and the binary quoting decision is robust to small weight perturbations.

12 Discussion

12.1 Interpretation of Results

The central empirical finding is that toxicity screening combined with inventory risk control is *necessary for viability*. The mechanism is loss avoidance rather than alpha generation: the strategy declines to quote when expected PnL is negative, avoiding the inventory accumulation that destroys the unscreened baseline.

The γ sensitivity cliff (Table 3) reflects a structural challenge: periodic order book cleanup causes visible queue depth to appear shallower than reality, inflating apparent fill rates. The inventory risk penalty offsets this by suppressing quoting as inventory accumulates. The weak per-fill toxicity signal ($\rho = 0.124$) succeeds because the E[PnL] filter integrates it with spread capture, rebates, and inventory risk to produce a binary quoting decision applied consistently across tens of thousands of opportunities per day. The SGD convergence result (Table 6) confirms cancellation features dominate and validates the initial weight assignment.

12.2 Limitations

Key limitations include: (1) *single-day sample*–results from August 22, 2023 may not generalize across volatility regimes; (2) *execution model simplifications*–negligible market impact, idealized queue positioning, parametric latency, and order book staleness from periodic cleanup of tracked orders (which inflates apparent fill rates); (3) γ *sensitivity*–the strategy is critically dependent on the inventory risk parameter, as a $4\times$ reduction from the optimal γ converts positive PnL to $-\$32$K losses; (4) *process-group independence*–initial hypothesis tests treat 14 file groups as independent; (5) *incomplete cost model*–financing, opportunity, and infrastructure costs are excluded.

12.3 Future Work

The most immediate extension is multi-day validation across diverse market regimes to establish external validity. The online learning results suggest that further toxicity model improvement requires either longer lookforward horizons (the 250μs window may be too short for temporal features to carry signal) or cross-asset signals (correlated ETF flow, index futures activity) that capture information arrival before it manifests in single-stock microstructure. Additional extensions include reinforcement learning policy optimization, walk-forward parameter calibration, and endogenous market impact modeling.

13 Conclusion

We extend the Avellaneda-Stoikov framework with real-time adverse selection screening via a microstructure-based toxicity score. The toxicity score is formalized as a logistic function of 15 microstructure features (5 order book ratios, 3 temporal signals, 7 structural features) with SGD-trained weights, integrated into an expected PnL filter with quadratic inventory risk penalty for optimal quoting decisions–a framework independent of the specific dataset. A C++ simulation processing 1.27 billion messages at 5.85M msgs/sec via hybrid multi-process architecture with memory-mapped I/O enables large-scale microstructure research on commodity hardware. Empirically, a weak per-fill toxicity signal ($\rho = 0.124$, $n = 30{,}816$) aggregates into meaningful portfolio improvement via the expected PnL filter: all three filter variants achieve positive PnL (SGD logistic +\$1,266, walk-forward +\$1,353, EWMA +\$277), and SGD weight convergence confirms cancellation features are the dominant predictors (cancel ratio $\rho = 0.101$, cancel volume intensity $\rho = 0.123$).

The E[PnL] filter must use base spread rather than toxicity-adjusted spread, because fills near the NBBO capture base spread regardless of quoted spread width. The quadratic inventory penalty γQ^2 is the critical mechanism, with $\gamma = 2 \times 10^{-7}$ as the precise sweet spot; this naturally suppresses quoting as inventory accumulates, preventing the one-sided exposure that destroys the unscreened baseline.

Loss avoidance through selective quoting and inventory risk control is the dominant mechanism. The framework provides a complete, implementable system for evaluating toxicity-screened market-making strategies on large-scale market data.

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